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Carleson measures, multipliers and integration operators for spaces of Dirichlet type

Daniel Girela [a,*], José Ángel Peláez [b]

[a] *Departamento de Análisis Matemático, Facultad de Ciencias, Universidad de Málaga, Campus de Teatinos, 29071 Málaga, Spain*
[b] *Departamento de Análisis Matemático, Universidad de Sevilla, Avenida de la Reina Mercedes, Apartado de Correos 1160, 41080 Sevilla, Spain*

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Abstract

For $0 < p < \infty$ and $\alpha > -1$, we let $\mathcal{D}^p_\alpha$ denote the space of those functions f which are analytic in the unit disc $\mathbb{D} = \{z \in \mathbb{C}: |z| < 1\}$ and satisfy $\int_{\mathbb{D}} (1-|z|^2)^\alpha |f'(z)|^p \, dx\, dy < \infty$. In this paper we characterize the positive Borel measures μ in $\mathbb{D}$ such that $\mathcal{D}^p_\alpha \subset L^q(d\mu)$, $0 < p < q < \infty$. We also characterize the pointwise multipliers from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ $(0 < p < q < \infty)$ if $p - 2 < \alpha < p$. In particular, we prove that if $(2+\alpha)/p - (\beta+2)/q > 0$ the only pointwise multiplier from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ $(0 < p < q < \infty)$ is the trivial one. This is not longer true for $(2+\alpha)/p - (\beta+2)/q \leqslant 0$ and we give a number of explicit examples of functions which are multipliers from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ for this range of values.
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1. Introduction and main results

Let $\mathbb{D}$ denote the open unit disc of the complex plane $\mathbb{C}$ and $\mathcal{H}ol(\mathbb{D})$ the space of all analytic functions in $\mathbb{D}$. If $0 < r < 1$ and f is an analytic function in $\mathbb{D}$ we set

* Corresponding author.
E-mail addresses: girela@uma.es (D. Girela), japelaez@us.es (J.Á. Peláez).

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$$M_p(r,f)=\left(\frac{1}{2\pi}\int_0^{2\pi}\left|f\left(re^{it}\right)\right|^p dt\right)^{1/p},\qquad I_p(r,f)=M_p^p(r,f)\quad (0<p<\infty),$$

$$M_\infty(r,f)=\sup_{0\leqslant t\leqslant 2\pi}\left|f\left(re^{it}\right)\right|.$$

For $0<p<\infty$ and $\alpha>-1$, the Bergman space A^p_α is the set of all $f\in\mathcal{H}ol(\mathbb{D})$ for which

$$\|f\|_{A^p_\alpha}\stackrel{\text{def}}{=}\left(\int_{\mathbb{D}}\left|f(z)\right|^p\left(1-|z|\right)^\alpha dA(z)\right)^{1/p}<\infty,$$

where $dA(z)=\frac{1}{\pi}\,dx\,dy$ is the normalized Lebesgue area measure. We mention [10,17] as general references for the theory of Bergman spaces.

If $0<p<\infty$ and $\alpha>-1$, the space of Dirichlet type $\mathcal{D}^p_\alpha$ consists of those functions f which are analytic in $\mathbb{D}$ such that

$$\|f\|_{\mathcal{D}^p_\alpha}\stackrel{\text{def}}{=}\left(\left|f(0)\right|^p+\int_{\mathbb{D}}\left(1-|z|\right)^\alpha\left|f'(z)\right|^p dA(z)\right)^{1/p}<\infty.$$

That is, $f\in\mathcal{D}^p_\alpha$ if and only if $f'\in A^p_\alpha$.

If $I\subset\partial\mathbb{D}$ is an interval of $\partial\mathbb{D}$, $|I|$ will denote the length of I. The *Carleson square* $S(I)$ is defined as

$$S(I)=\left\{re^{it}\colon\ e^{it}\in I,\ 1-\frac{|I|}{2\pi}\leqslant r<1\right\}.$$

Also, for $a\in\mathbb{D}$, $S(a)$ will denote the *Carleson box* with vertex at a, that is,

$$S(a)=\left\{z\in\mathbb{D}\colon\ 1-|z|\leqslant 1-|a|,\ \left|\frac{\arg(a\bar{z})}{2\pi}\right|\leqslant\frac{1-|a|}{2}\right\}.$$

If $s>0$ and μ is a positive Borel measure in $\mathbb{D}$, we shall say that μ is an s-Carleson measure if there exists a positive constant C such that

$$\mu\left(S(I)\right)\leqslant C|I|^s\quad\text{for any interval } I\subset\partial\mathbb{D},\tag{1}$$

or, equivalently, if there exists $C>0$ such that

$$\mu\left(S(a)\right)\leqslant C\left(1-|a|\right)^s\quad\text{for all } a\in\mathbb{D}.\tag{2}$$

An 1-Carleson measure will be simply called a Carleson measure.

For a large class of spaces X of analytic functions in $\mathbb{D}$, a characterization of those positive Borel measures μ in $\mathbb{D}$ such that $X\subset L^p(d\mu)$ is known and such a characterization is useful to study the boundedness of operators acting on X. We recall that Carleson [7] proved that $H^p\subset L^p(d\mu)$, $0<p<\infty$, if and only if μ is a Carleson measure. This result was extended by Duren [8] (see also [9, Theorem 9.4]) who proved that for $0<p\leqslant q<\infty$ $H^p\subset L^q(d\mu)$ if and

only if μ is a q/p-Carleson measure. Luecking [20], characterized those positive Borel measures μ in $\mathbb{D}$ for which $A^p_\alpha \subset L^q(d\mu)$, $0 < p \leqslant q < \infty$. Indeed, the following result is contained in [20, Theorem 2.2].

Theorem A. *Suppose that $0 < p \leqslant q < \infty$, $-1 < \alpha < \infty$ and let μ be a positive Borel measure in $\mathbb{D}$. Then $A^p_\alpha \subset L^q(d\mu)$ if and only if μ is a $\frac{(2+\alpha)q}{p}$-Carleson measure.*

In this paper we shall be interested in characterizing the measures μ for which $\mathcal{D}^p_\alpha \subset L^q(d\mu)$, $0 < p \leqslant q < \infty$. This question has been studied for certain values of p, q and α, among other, by the following authors: Arcozzi, Rochberg and Sawyer [4], Verbitsky [24], Vinogradov [25], and Z. Wu [26].

Vinogradov [25] and Z. Wu [26], independently, have proved the following result.

Theorem B. *Suppose that $0 < p \leqslant 2$ and let μ be a positive Borel measure in $\mathbb{D}$, then $\mathcal{D}^p_{p-1} \subset L^p(d\mu)$ if and only if μ is an 1-Carleson measure.*

Z. Wu [26, Theorem 1] also obtained a characterization of those μ such that $\mathcal{D}^p_\alpha \subset L^p(d\mu)$ for all p, α with $\alpha > -1$ and $\alpha + 1 < p$. He left open the case $p = \alpha + 1 > 2$ but conjectured that his result for $1 < p = \alpha + 1 \leqslant 2$ could be extended to this range, that is, he conjectured that Theorem B remains true for $2 < p < \infty$. Recently, we have proved that Wu's conjecture is false. Indeed, the following result is contained in [15, Theorem 2.1].

Theorem C. *Suppose that $2 < p < \infty$. Then there exists an 1-Carleson measure μ in $\mathbb{D}$, such that $\mathcal{D}^p_{p-1} \nsubseteq L^p(d\mu)$.*

Let us remark that the inclusion

$$\mathcal{D}^p_{p-1} \subset H^p, \quad 0 < p \leqslant 2 \tag{3}$$

(see [25, Lemma 1.4]) plays an essential role in the proof of Theorem B. A classical result of Littlewood and Paley [19] (see also [21]) asserts that

$$H^p \subset \mathcal{D}^p_{p-1}, \quad 2 \leqslant p < \infty. \tag{4}$$

Bearing in mind Duren's characterization of those μ for which $H^p \subset L^q(d\mu)$ $(p < q)$, (3), (4), Theorems B and C, it is natural to make the following conjectures:

Conjecture 1. *If $0 < p \leqslant 2$, $p < q$ and μ is a positive Borel measure in $\mathbb{D}$, then $\mathcal{D}^p_{p-1} \subset L^q(d\mu)$ if and only if μ is a q/p-Carleson measure.*

Conjecture 2. *If $2 < p < \infty$ and $p < q$ then there exists a q/p-Carleson measure μ in $\mathbb{D}$ such that $\mathcal{D}^p_{p-1} \nsubseteq L^q(d\mu)$.*

In Theorem 1, for any given p, q, α with $0 < p < q < \infty$ and $\alpha > -1$, we give a characterization of those positive Borel measures μ in $\mathbb{D}$ for which $\mathcal{D}^p_\alpha \subset L^q(\mu)$.

Theorem 1. *Suppose that $0 < p < q < \infty$ and $\alpha > -1$ and let μ be a positive Borel measure in $\mathbb{D}$.*

(a) *If $p < \alpha + 2$, then $\mathcal{D}^p_\alpha \subset L^q(\mu)$ if and only if μ is a $\frac{q}{p}(\alpha - p + 2)$-Carleson measure.*

(b) *If $p = \alpha + 2$, then $\mathcal{D}^p_\alpha \subset L^q(\mu)$ if and only if there exists a positive constant C such that*

$$\mu\big(S(I)\big) \leqslant C\left(\left(\log \frac{1}{|I|}\right)^{(\frac{1}{p}-1)q}\right) \quad \text{for all intervals } I \subset \partial\mathbb{D}. \tag{5}$$

(c) *If $p > \alpha + 2$, then $\mathcal{D}^p_\alpha \subset L^q(\mu)$ if and only if μ is a finite measure.*

We note that this implies that Conjecture 1 is true but Conjecture 2 is false. We remark also that Theorem 1 is essentially known for some values of p and α. Indeed, we shall see that:

- for $0 < p < \alpha + 1$, it is an easy consequence of Luecking's Theorem A;
- for $1 < p$ and $\alpha + 1 < p \leqslant \alpha + 2$, it follows from the work of Arcozzi et al. [4];
- for $p > \alpha + 2$, it follows from the inclusion $\mathcal{D}^p_\alpha \subset H^\infty$ which is proved in the course of the proof of [26, Theorem 4.2].

Now we turn to study the boundedness of several operators from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$, $0 < p < q$, $\alpha, \beta > -1$.

For g analytic in $\mathbb{D}$, the integration operator I_g is defined as follows:

$$I_g(f)(z) \stackrel{\text{def}}{=} \int_0^z g(\xi) f'(\xi)\, d\xi, \quad f \in \mathcal{H}ol(\mathbb{D}),\ z \in \mathbb{D}. \tag{6}$$

The operators I_g have been studied in a number of papers and contain as special cases a number of important operators such as the integral operator (I_g with $g(z) = z$) and the Cesàro operator (I_g with $g(z) = \log(1/(1-z))$). Let us mention that Aleman and Cima characterized in [1] those $g \in \mathcal{H}ol(\mathbb{D})$ for which I_g maps H^p into H^q and Aleman, Siskakis studied in [2] the operators I_g acting on Bergman spaces.

The multiplication operator M_g is defined by

$$M_g(f)(z) \stackrel{\text{def}}{=} g(z) f(z), \quad f \in \mathcal{H}ol(\mathbb{D}),\ z \in \mathbb{D}. \tag{7}$$

We shall characterize the boundedness of the above operators from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$, $0 < p < q$. Indeed, we shall prove the following results in Section 3.

Theorem 2. *Suppose that $g \in \mathcal{H}ol(\mathbb{D})$, $0 < p < q$ and $\alpha, \beta > -1$. If $p > \alpha$ then the following conditions are equivalent*:

(i) *I_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$.*

(ii) $M_\infty(r, g) = \mathrm{O}((1-r)^{\frac{2+\alpha}{p} - \frac{\beta+2}{q}})$.

Theorem 3. *Suppose that $g \in \mathcal{H}ol(\mathbb{D})$, $0 < p < q$ and $\alpha, \beta > -1$. If $p - 2 < \alpha < p$, the following conditions are equivalent*:

(i) *M_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$.*

(ii) *The positive Borel measure* $\mu_{g,q,\beta}$ *in* $\mathbb{D}$ *defined by*

$$d\mu_{g,q,\beta}(z) = \left(1-|z|^2\right)^{\beta} \left|g'(z)\right|^q dA(z)$$

is a $\frac{q}{p}(\alpha - p + 2)$*-Carleson measure and*

$$M_\infty(r, g) = \mathrm{O}\left((1-r)^{\frac{2+\alpha}{p} - \frac{\beta+2}{q}}\right).$$

As an immediate consequence of Theorem 3 we deduce the following result.

Corollary 1. *Suppose that* $g \in \mathcal{H}ol(\mathbb{D})$, $0 < p < q$ *and* $\alpha, \beta > -1$. *If* $p - 2 < \alpha < p$, *and* $\frac{2+\alpha}{p} - \frac{\beta+2}{q} > 0$, *the following conditions are equivalent*:

(i) $I_g : \mathcal{D}^p_\alpha \to \mathcal{D}^q_\beta$ *is a bounded operator.*
(ii) $M_g : \mathcal{D}^p_\alpha \to \mathcal{D}^q_\beta$ *is a bounded operator.*
(iii) $g \equiv 0$.

If X and Y are two spaces of analytic functions in $\mathbb{D}$ and $g \in \mathcal{H}ol(\mathbb{D})$, then g is said to be pointwise multiplier from X to Y if $M_g(X) \subset Y$. The space of all pointwise multipliers from X to Y will be denoted by $M(X, Y)$. Since $\mathcal{D}^{p_2}_\alpha \subset \mathcal{D}^{p_1}_\alpha$, $0 < p_1 \leqslant p_2 < \infty$, we trivially deduce the following.

Lemma 1. *Suppose that* $\alpha, \beta > -1$, $0 < p_1 \leqslant p_2 < \infty$ *and* $0 < q < \infty$. *Then* $M(\mathcal{D}^{p_1}_\alpha, \mathcal{D}^q_\beta) \subset M(\mathcal{D}^{p_2}_\alpha, \mathcal{D}^q_\beta)$.

Corollary 1 asserts that if $\frac{2+\alpha}{p} - \frac{\beta+2}{q} > 0$ the only pointwise multiplier from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ ($0 < p < q < \infty$) is the trivial one. This is not longer true for $\frac{2+\alpha}{p} - \frac{\beta+2}{q} \leqslant 0$. In Section 4 we shall give a number of explicit examples of functions belonging to $M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$ for this range of values.

Next we shall get into the proofs of these and some other results but, before doing so, let us remark that, as usual, we shall be using the convention that $C_{p,\alpha,q,\beta,\ldots}$ will denote a positive constant which depends only upon the displayed parameters $p, \alpha, q, \beta, \ldots$ but not necessarily the same at different occurrences. Also, in many cases we shall omit parameters.

2. Proof of Theorem 1

We shall split the proof of Theorem 1 in five cases. Cases A, B, and C are those in which, as mentioned in Section 1, the result is essentially known.

Case A *(*$0 < p < \alpha + 1$*).* Flett [11, Theorem 6] proved that $\mathcal{D}^p_\alpha = A^p_{\alpha-p}$ if $0 < p < \alpha + 1$. Then the result follows using Luecking's Theorem A.

Case B *($p > 1$ and $\alpha + 1 < p \leqslant \alpha + 2$).* If $1 < p < \infty$ and ϱ is a positive Borel measurable weight function in $\mathbb{D}$, the weighted Besov space $B_p(\varrho)$ is defined as the space of those $f \in \mathcal{H}ol(\mathbb{D})$ such that

$$\|f\|^p_{B_p(\varrho)} = \big|f(0)\big|^p + \int_{\mathbb{D}} \big(1-|z|^2\big)^p \big|f'(z)\big|^p \varrho(z) \frac{dA(z)}{(1-|z|^2)^2} < \infty.$$

We have $\mathcal{D}^p_\alpha = B_p((1-|z|)^{\alpha-p+2})$. Furthermore, following the terminology of [4, p. 445], $\varrho(z) = ((1-|z|)^{\alpha-p+2})$ is a *p-admissible* weight if and only if $-1 < \alpha < p-1$. Arcozzi et al. [4] proved the following result.

Theorem D. *Suppose that $1 < p < q < \infty$ and that ϱ is a p-admissible weight. Let μ be a positive Borel measure in $\mathbb{D}$, then the following conditions are equivalent*:

(i) *There exists a positive constant $C(\mu)$ such that*

$$\|f\|_{L^q(d\mu)} \leqslant C(\mu)\|f\|_{B_p(\varrho)} \quad \text{for all } f \in B_p(\varrho). \tag{8}$$

(ii) *There exists a positive constant $C_2(\mu)$ such that*

$$\mu\big(S(a)\big) \leqslant C_2(\mu)\bigg(\int_{[0,a]} \varrho(w)^{1-p'} \frac{|dw|}{1-|w|^2}\bigg)^{-q/p'} \quad \text{for all } a \in \mathbb{D}. \tag{9}$$

Here and throughout the paper

$$\frac{1}{p} + \frac{1}{p'} = \frac{1}{q} + \frac{1}{q'} = 1 \quad (p, q > 1).$$

It is not difficult to prove the following result.

Lemma 2. *Let $\varrho(w) = ((1-|w|)^{\alpha-p+2})$, $w \in \mathbb{D}$.*

(i) *If $p > 1$ and $\alpha > p-2$, then*

$$\bigg(\int_{[0,a]} \varrho(w)^{1-p'} \frac{|dw|}{1-|w|^2}\bigg)^{-q/p'} \asymp \big(1-|a|\big)^{\frac{q}{p}(\alpha-p+2)} \quad \text{as } |a| \to 1^-.$$

(ii) *If $p > 1$ and $\alpha = p-2$, then*

$$\bigg(\int_{[0,a]} \varrho(w)^{1-p'} \frac{|dw|}{1-|w|^2}\bigg)^{-q/p'} \asymp \bigg(\log \frac{1}{1-|a|}\bigg)^{-q/p'} \quad \text{as } |a| \to 1^-.$$

Theorem D and Lemma 2 easily yield the following.

Proposition 1. *Suppose that $1 < p < q < \infty$ and let μ be a positive Borel measure in $\mathbb{D}$.*

(i) *If $\alpha + 1 < p < \alpha + 2$, then $\mathcal{D}^p_\alpha \subset L^q(d\mu)$ if and only if μ is a $\frac{q}{p}(\alpha - p + 2)$-Carleson measure.*
(ii) *If $p = \alpha + 2$, then $\mathcal{D}^p_\alpha \subset L^q(d\mu)$ if and only if there exists a positive constant C such that*

$$\mu\big(S(I)\big) \leqslant C\left(\left(\log\frac{1}{|I|}\right)^{(\frac{1}{p}-1)q}\right) \quad \text{for all intervals } I \subset \partial\mathbb{D}.$$

Hence the proof in this case is finished.

Case C *($p > \alpha + 2$).* If $p > \alpha + 2$, then $\mathcal{D}^p_\alpha \subset H^\infty$ (see [26]) and then it follows easily that $\mathcal{D}^p_\alpha \subset L^q(d\mu)$ if and only if μ is a finite measure. Indeed, if μ is a finite measure it is clear that $H^\infty \subset L^q(d\mu)$ and, hence, $\mathcal{D}^p_\alpha \subset L^q(d\mu)$. Conversely, assuming that $\mathcal{D}^p_\alpha \subset L^q(d\mu)$ the fact that μ is a finite measures follows using that the function identically equal to 1 belongs to $\mathcal{D}^p_\alpha$.

Case D *($p = \alpha + 1$).* In this case Theorem 1 reduces to the following.

Theorem 4. *Let $0 < p < q < \infty$ and let μ be a positive Borel measure in $\mathbb{D}$. Then the following conditions are equivalent*:

(i) $\mathcal{D}^p_{p-1} \subset L^q(d\mu)$.
(ii) *μ is a $\frac{q}{p}$-Carleson measure.*

Several results will be needed in the proof. The following lemma is stated as Lemma 3.4 of [16], it is essentially due to Hardy and Littlewood and can be proved by modifying the proof of Theorem 5.9 in [9].

Lemma 3. *For $0 < s < t \leqslant \infty$, there exists a positive constant C_{st} depending only on s and t such that, for each $f \in \mathcal{H}ol(\mathbb{D})$ and each $r \in (0, 1)$,*

$$M_t(r, f) \leqslant C_{st} M_s\left(\frac{1+r}{2}, f\right)(1-r)^{\frac{1}{t}-\frac{1}{s}}. \tag{10}$$

Proposition 2. *Let $0 < p < \infty$ and $\alpha > -1$, then $\mathcal{D}^p_{p-1} \subset A^{(2+\alpha)p}_\alpha$.*

Proof. Theorem 6 of [11] implies that

$$f \in A^s_\gamma \quad \Leftrightarrow \quad f' \in A^s_{s+\gamma}, \quad s > 0,\ \gamma > -1. \tag{11}$$

Take $f \in \mathcal{D}^p_{p-1}$. Using Lemma 3 with $s = p$ and $t = (2+\alpha)p$, we deduce that there exists a positive constant C such that

$$I_{(2+\alpha)p}(r, f') \leqslant C M_p^{(2+\alpha)p}\left(\frac{1+r}{2}, f'\right)(1-r)^{-1-\alpha}, \quad 0 < r < 1. \tag{12}$$

On the other hand, since $f \in \mathcal{D}^p_{p-1}$,

$$M_p(r, f') = o\left(\frac{1}{1-r}\right) \quad \text{as } r \to 1^-. \tag{13}$$

Bearing in mind (12), (13), the fact that $f \in \mathcal{D}^p_{p-1}$ and making a change of variables we deduce that

$$\begin{aligned}\int_0^1 I_{(2+\alpha)p}(r, f')(1-r)^{(2+\alpha)p+\alpha}\, dr &\leqslant C \int_0^1 M_p^{(2+\alpha)p}\left(\frac{1+r}{2}, f'\right)(1-r)^{(2+\alpha)p-1}\, dr \\ &\leqslant C \int_0^1 M_p^p(r, f') M_p^{p+\alpha p}(r, f')(1-r)^{(2+\alpha)p-1}\, dr \\ &\leqslant C \int_0^1 M_p^p(r, f')(1-r)^{p-1}\, dr < \infty.\end{aligned}$$

Thus, $f' \in A^{(2+\alpha)p}_{(2+\alpha)p+\alpha}$ which, using (11), implies that $f \in A^{(2+\alpha)p}_{\alpha}$. □

Lemma 4. *Let $0 < p, q < \infty$ and let μ be a positive Borel measure in $\mathbb{D}$ such that $\mathcal{D}^p_{p-1} \subset L^q(d\mu)$ then μ is a $\frac{q}{p}$-Carleson measure.*

Proof. In order to prove that μ is a $\frac{q}{p}$-Carleson measure, if suffices to show that

$$\sup_{a \in \mathbb{D}} \int_{\mathbb{D}} \left(\frac{1-|a|^2}{|1-\bar{a}z|^2}\right)^{q/p} d\mu(z) < \infty \tag{14}$$

(see [5, Lemma 2.1]).

Putting together the embedding $\mathcal{D}^p_{p-1} \subset L^q(d\mu)$ and the closed graph theorem, we can assert that there exists a positive constant C such that

$$\|f\|^q_{L^q(d\mu)} \leqslant C \|f\|^q_{\mathcal{D}^p_{p-1}} \quad \text{for all } f \in \mathcal{H}ol(\mathbb{D}). \tag{15}$$

For $a \in \mathbb{D}$, define

$$f_a(z) = \left(\frac{1-|a|^2}{(1-\bar{a}z)^2}\right)^{1/p}, \quad z \in \mathbb{D}.$$

Using (15), we deduce that

$$\int_{\mathbb{D}} \left(\frac{1-|a|^2}{|1-\bar{a}z|^2}\right)^{q/p} d\mu(z) \leqslant C\left(|f_a(0)|^p + \int_{\mathbb{D}} (1-|z|^2)^{p-1} |f_a'(z)|^p \, dA(z)\right)^{q/p}$$

$$\leqslant C(1-|a|^2)^{q/p}\left(1+\left(\frac{2|a|}{p}\right)^p \int_{\mathbb{D}} \frac{(1-|z|^2)^{p-1}}{|1-\bar{a}z|^{2+p}}\, dA(z)\right)^{q/p}$$

$$\leqslant C(1-|a|^2)^{q/p}\left(1+\left(\frac{2|a|}{p}\right)^p \int_0^1 \frac{(1-r^2)^{p-1}}{(1-|a|r)^{1+p}}\, dr\right)^{q/p}$$

$$\leqslant C(1-|a|^2)^{q/p}\left(1+\left(\frac{2|a|}{p}\right)^p \frac{1}{1-|a|^2}\right)^{q/p} \leqslant C < \infty.$$

This proves (14) and finishes the proof. □

Proof of Theorem 4. The implication (i) ⇒ (ii) follows from Lemma 4.

Suppose now that μ is a (q/p)-Carleson measure. Take $\alpha > -1$ such that $(2+\alpha)p < q$ (such an α exists because $p < q$). Proposition 2 implies that $\mathcal{D}^p_{p-1} \subset A^{(2+\alpha)p}_\alpha$. Now, Theorem A implies that $A^{(2+\alpha)p}_\alpha \subset L^q(d\mu)$ and, hence, $\mathcal{D}^p_{p-1} \subset L^q(d\mu)$. □

Case E *($p \leqslant 1$ and $\alpha + 1 < p$).* In this case Theorem 1 reduces to the following.

Theorem 5. *Suppose that $0 < p \leqslant 1$, $-1 < \alpha < p-1$ and $p < q < \infty$. Let μ be a positive Borel measure in $\mathbb{D}$. Then the following conditions are equivalent*:

(i) $\mathcal{D}^p_\alpha \subset L^q(d\mu)$.
(ii) μ *is a* $\frac{q}{p}(\alpha - p + 2)$*-Carleson measure.*

The following result will be used in the proof of Theorem 5. It is a particular case of the first part of [23, Proposition 2.1].

Proposition A. *Let s, t, s_1 and s_2 be positive numbers such that $s = s_1 \cdot s_2$ and $s_1 < t$. Let μ be a positive Borel measure in $\mathbb{D}$. Then μ is an s-Carleson measure if and only if*

$$\sup_{a\in\mathbb{D}} \int_{\mathbb{D}} \left(\frac{(1-|a|^2)^{t-s_1}}{|1-\bar{a}z|^t}\right)^{s_2} d\mu(z) < \infty. \tag{16}$$

Proof of Theorem 5. Let p, q and α be as in Theorem 5.

Suppose first that $\mathcal{D}^p_\alpha \subset L^q(d\mu)$. Then, by the closed graph theorem, there exists a positive constant C such that

$$\|f\|_{L^q(d\mu)} \leqslant C\|f\|_{\mathcal{D}^p_\alpha}. \tag{17}$$

Applying (17) to the test function $f_a(z) = (1-\bar{a}z)^{-2/p}$ and bearing in mind that $p > \alpha + 1 > \alpha$, we deduce that

$$\int_{\mathbb{D}} \left(\frac{1}{|1-\overline{a}z|^2} \right)^{q/p} d\mu(z) \leqslant C \left(1 + \left(\frac{2|a|}{p} \right)^p \int_{\mathbb{D}} \frac{(1-|z|^2)^\alpha}{|1-\overline{a}z|^{2+p}}\, dA(z) \right)^{q/p}$$

$$\leqslant C \left(1 + \frac{|a|^p}{(1-|a|^2)^{p-\alpha}} \right)^{q/p}.$$

Hence,

$$\sup_{a\in\mathbb{D}} \int_{\mathbb{D}} \left(\frac{(1-|a|^2)^{p-\alpha}}{|1-\overline{a}z|^2} \right)^{q/p} d\mu(z) < \infty. \tag{18}$$

Using Proposition A with $s = \frac{q}{p}(\alpha - p + 2)$, $s_1 = (\alpha - p + 2)$ and $t = 2$, we obtain that μ is a $\frac{q}{p}(\alpha - p + 2)$-Carleson measure.

Suppose now that μ is a $\frac{q}{p}(\alpha - p + 2)$-Carleson measure, then Proposition A shows that μ satisfies (18). Let $g \in \mathcal{D}^p_\alpha$ and assume, without loss of generality that $g(0) = 0$. Arguing as in the proof of [26, Theorem 3.3], we deduce that there exists $C > 0$ (independent of g) such that

$$\left|g(z)\right|^p \leqslant C \int_{\mathbb{D}} \frac{|g'(w)|^p}{|1-\overline{w}z|^2} \left(1-|w|^2\right)^p dA(w), \quad z \in \mathbb{D}.$$

Since $q/p > 1$, using Minkowski's inequality in continuous form and (18), we deduce that

$$\begin{aligned}
\left(\int_{\mathbb{D}} \left|g(z)\right|^q d\mu(z) \right)^{p/q} &\leqslant C \left(\int_{\mathbb{D}} \left(\int_{\mathbb{D}} \frac{|g'(w)|^p}{|1-\overline{w}z|^2} \left(1-|w|^2\right)^p dA(w) \right)^{q/p} d\mu(z) \right)^{p/q} \\
&\leqslant C \int_{\mathbb{D}} \left(\int_{\mathbb{D}} \frac{|g'(w)|^q}{|1-\overline{w}z|^{2q/p}} \left(1-|w|^2\right)^q d\mu(z) \right)^{p/q} dA(w) \\
&\leqslant C \int_{\mathbb{D}} \left|g'(w)\right|^p \left(1-|w|^2\right)^p \left(\int_{\mathbb{D}} \frac{d\mu(z)}{|1-\overline{w}z|^{2q/p}} \right)^{p/q} dA(w) \\
&\leqslant C \int_{\mathbb{D}} \left|g'(w)\right|^p \left(1-|w|^2\right)^p \left(1-|w|^2\right)^{\alpha-p} dA(w) \\
&\leqslant C \int_{\mathbb{D}} \left|g'(w)\right|^p \left(1-|w|^2\right)^\alpha dA(w).
\end{aligned}$$

This finishes the proof. □

3. Multipliers and integral operators

Let us start introducing another operator. If $g \in \mathcal{H}ol(\mathbb{D})$ the operator J_g is defined as follows:

$$J_g(f)(z) \stackrel{\text{def}}{=} \int_0^z g'(\xi) f(\xi)\, d\xi, \quad f \in \mathcal{H}ol(\mathbb{D}),\ z \in \mathbb{D}. \tag{19}$$

The importance of the operator J_g comes from the fact that

$$I_g(f) + J_g(f) = M_g(f) - f(0)g(0).$$

The following theorem follows directly from Theorem 1 and the closed graph theorem.

Theorem 6. *Suppose that $g \in \mathcal{H}ol(\mathbb{D})$, $0 < p < q$ and $\alpha, \beta > -1$. If $\alpha > p - 2$ the following conditions are equivalent*:

(i) *J_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$.*
(ii) *The positive Borel measure $\mu_{g,q,\beta}$ in $\mathbb{D}$ defined by*

$$d\mu_{g,q,\beta}(z) = \left(1 - |z|^2\right)^\beta \left|g'(z)\right|^q dA(z)$$

is a $\frac{q}{p}(\alpha - p + 2)$-Carleson measure.

Next we shall introduce some notation and collect some results concerning the pseudo-hyperbolic metric which will be needed in our work. The pseudo-hyperbolic metric in the unit disc will be denoted by ϱ and is defined as follows:

$$\varrho(z, w) = \left|\frac{z - w}{1 - \overline{w}z}\right|, \quad z, w \in \mathbb{D}.$$

A *pseudo-hyperbolic disc* of (pseudo-hyperbolic) center a and radius r ($a \in \mathbb{D}$, $0 < r < 1$) is the set $\Delta(a, r) = \{z \in \mathbb{D}\colon\ \varrho(a, z) < r\}$. It coincides with the Euclidean disc whose (Euclidean) radius and center are (see [12, p. 3] or [10, p. 40]):

$$R = \frac{1 - |a|^2}{1 - r^2|a|^2} r, \qquad c = \frac{1 - r^2}{1 - r^2|a|^2} a. \tag{20}$$

The following lemma is well known (see, e.g., [10, Section 2.5]).

Lemma A. *Let $0 < r < 1$. Then there exist positive constants C_1, C_2, C_3 and C_4 which depend only on r such that, if $a \in \mathbb{D}$ and $z \in \Delta(a, r)$, then*

$$1 - |a|^2 \leqslant C_1\left(1 - |z|^2\right) \leqslant C_2|1 - \bar{a}z| \leqslant C_3\left|\Delta(a, r)\right|^{1/2} \leqslant C_4\left(1 - |a|^2\right). \tag{21}$$

Here, $|\Delta(a, r)|$ denotes the Lebesgue area measure of $\Delta(a, r)$.

Proof of Theorem 2. (i) $\Rightarrow$ (ii). Let $g \in \mathcal{H}ol(\mathbb{D})$ and suppose that I_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$. Take and fix $r \in (0,1)$. It is well known (see, e.g., [27, Proposition 4.3.8], or [10, Lemma 13, p. 63]) that there exists a positive constant C such that

$$\big|g(a)\big|^q \leqslant \frac{C}{|\Delta(a,r)|} \int\limits_{\Delta(a,r)} \big|g(z)\big|^q \, dA(z), \quad a \in \mathbb{D}. \tag{22}$$

For each $a \in \mathbb{D}$, set

$$f_a(z) = \left(\frac{(1-|a|^2)^{p-\alpha}}{(1-\bar{a}z)^2} \right)^{1/p}, \quad z \in \mathbb{D}.$$

We have

$$\|f_a\|^p_{\mathcal{D}^p_\alpha} \leqslant 1 + C\big(1-|a|^2\big)^{p-\alpha} \int\limits_{\mathbb{D}} \frac{(1-|z|^2)^\alpha}{|1-\bar{a}z|^{p+2}} \, dA(z), \quad a \in \mathbb{D}.$$

Then, using [27, Lemma 4.2.2] with $t = \alpha$ and $c = p - \alpha > 0$, we deduce that

$$K \stackrel{\text{def}}{=} \sup_{a \in \mathbb{D}} \|f_a\|_{\mathcal{D}^p_\alpha} < \infty. \tag{23}$$

Using Lemma A, (22) and (23), we obtain

$$\begin{aligned}
|a|^q \big(1-|a|^2\big)^{\beta+2-\frac{q(2+\alpha)}{p}} \big|g(a)\big|^q &\leqslant C|a|^q \big(1-|a|^2\big)^{\beta-\frac{q(2+\alpha)}{p}} \int\limits_{\Delta(a,r)} \big|g(z)\big|^q \, dA(z) \\
&\leqslant C|a|^q \int\limits_{\Delta(a,r)} \big|g(z)\big|^q \big(1-|z|^2\big)^{\beta-\frac{q(2+\alpha)}{p}} \, dA(z) \\
&\leqslant C|a|^q \int\limits_{\Delta(a,r)} \big|g(z) f_a'(z)\big|^q \big(1-|z|^2\big)^{\beta} \, dA(z) \\
&\leqslant C \int\limits_{\Delta(a,r)} \big|I_g(f_a)'(z)\big|^q \big(1-|z|^2\big)^{\beta} \, dA(z) \\
&\leqslant C \big\|I_g(f_a)\big\|^q_{\mathcal{D}^q_\beta} \\
&\leqslant C \|I_g\|^q_{(\mathcal{D}^p_\alpha \to \mathcal{D}^q_\beta)} \|f_a\|^q_{\mathcal{D}^p_\alpha} \\
&\leqslant C K^q \|I_g\|^q_{(\mathcal{D}^p_\alpha \to \mathcal{D}^q_\beta)}.
\end{aligned} \tag{24}$$

Thus

$$|a|^q \big|g(a)\big|^q \leqslant C \|I_g\|^q_{(\mathcal{D}^p_\alpha \to \mathcal{D}^q_\beta)} \big(1-|a|^2\big)^{-\beta-2+\frac{q(2+\alpha)}{p}}, \quad a \in \mathbb{D},$$

which implies that

$$M_\infty(r,g)=\mathrm{O}\big((1-r)^{\frac{2+\alpha}{p}-\frac{\beta+2}{q}}\big). \tag{25}$$

(ii) $\Rightarrow$ (i). Suppose that $g\in\mathcal{H}ol(\mathbb{D})$ satisfies (25). Take $f\in\mathcal{D}^p_\alpha$. It is easy to see that there exists a positive constant C (which does not depend on f) such that

$$\big|f'(z)\big|\leqslant C\|f\|_{\mathcal{D}^p_\alpha}\big(1-|z|^2\big)^{-\frac{\alpha+2}{p}},\quad z\in\mathbb{D}.$$

Thus

$$\begin{aligned}
\big\|I_g(f)\big\|^q_{\mathcal{D}^q_\beta}&=\int_{\mathbb{D}}\big|f'(z)g(z)\big|^q\big(1-|z|^2\big)^\beta\,dA(z)\\
&\leqslant C\int_{\mathbb{D}}\big|f'(z)\big|^p\big|f'(z)\big|^{q-p}\big(1-|z|^2\big)^{\frac{q(2+\alpha)}{p}-\beta-2}\big(1-|z|^2\big)^\beta\,dA(z)\\
&\leqslant C\|f\|^{q-p}_{\mathcal{D}^p_\alpha}\int_{\mathbb{D}}\big|f'(z)\big|^p\frac{1}{(1-|z|^2)^{\frac{(q-p)(\alpha+2)}{p}}}\big(1-|z|^2\big)^{\frac{q(2+\alpha)}{p}-2}\,dA(z)\\
&\leqslant C\|f\|^{q-p}_{\mathcal{D}^p_\alpha}\int_{\mathbb{D}}\big|f'(z)\big|^p\big(1-|z|^2\big)^\alpha\,dA(z)\\
&\leqslant C\|f\|^q_{\mathcal{D}^p_\alpha},
\end{aligned}$$

that is, I_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$. This finishes the proof. □

Proof of Theorem 3. The implication (ii) $\Rightarrow$ (i) follows trivially from Theorems 6 and 2, using the fact that $M_g(f)'=J_g(f)'+I_g(f)'$.

(i) $\Rightarrow$ (ii). Let $g\in\mathcal{H}ol(\mathbb{D})$ and suppose that M_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$. First, using an argument which is similar to that in the proof of (i) $\Rightarrow$ (ii) in Theorem 2, we shall prove that

$$M_\infty(r,g)=\mathrm{O}\big((1-r)^{\frac{2+\alpha}{p}-\frac{\beta+2}{q}}\big). \tag{26}$$

Take and fix $r\in(0,1)$. For each $a\in\mathbb{D}$, set

$$\tilde{f}_a(z)=\left(\frac{(1-|a|^2)^{p-\alpha}}{(1-\bar{a}z)^2}\right)^{1/p}-\big(1-|a|^2\big)^{1-\frac{\alpha}{p}},\quad z\in\mathbb{D}.$$

Since $\tilde{f}_a'=f_a'$, (23) implies that

$$\widetilde{K}\stackrel{\mathrm{def}}{=}\sup_{a\in\mathbb{D}}\|\tilde{f}_a\|_{\mathcal{D}^p_\alpha}<\infty. \tag{27}$$

Bearing in mind that $\tilde{f}_a(a)=0$ and using Lemma A, (22) with $M_g(\tilde{f}_a)$ in the place of g, and (27), we deduce that, for every $a\in\mathbb{D}$,

$$
\begin{aligned}
|a|^q\big(1-|a|^2\big)^{\beta+2-\frac{q(2+\alpha)}{p}}\big|g(a)\big|^q &\leqslant C|a|^q\big(1-|a|^2\big)^{\beta+2}\big|g(a)\tilde{f}_a'(a)\big|^q\\
&\leqslant C\big(1-|a|^2\big)^{\beta+2}\big|g(a)\tilde{f}_a'(a)+g'(a)\tilde{f}_a(a)\big|^q\\
&\leqslant C\big(1-|a|^2\big)^{\beta+2}\big|M_g(\tilde{f}_a)'(a)\big|^q\\
&\leqslant C\big(1-|a|^2\big)^{\beta}\int\limits_{\Delta(a,r)}\big|M_g(\tilde{f}_a)'(z)\big|^q\,dA(z)\\
&\leqslant C\int\limits_{\Delta(a,r)}\big|M_g(\tilde{f}_a)'(z)\big|^q\big(1-|z|^2\big)^{\beta}\,dA(z)\\
&\leqslant C\int\limits_{\mathbb{D}}\big|M_g(\tilde{f}_a)'(z)\big|^q\big(1-|z|^2\big)^{\beta}\,dA(z)\\
&\leqslant C\big\|M_g(\tilde{f}_a)\big\|^q_{\mathcal{D}^q_\beta}\\
&\leqslant C\|M_g\|^q_{(\mathcal{D}^p_\alpha\to\mathcal{D}^q_\beta)}\|\tilde{f}_a\|^q_{\mathcal{D}^p_\alpha}\\
&\leqslant C\widetilde{K}^q\|M_g\|^q_{(\mathcal{D}^p_\alpha\to\mathcal{D}^q_\beta)}.
\end{aligned}
$$

Thus, g satisfies (26). Now, Theorem 2 shows that I_g is continuous from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$. Since $J_g(f)'=M_g(f)'-I_g(f)'$, for all f, it follows that J_g is continuous from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ also. Then, Theorem 6 implies that the measure $\mu_{g,q,\beta}$ is a $\frac{q}{p}(\alpha-p+2)$-Carleson measure. This finishes the proof. □

Theorem 3 becomes simpler in the case $\frac{q(2+\alpha-p)}{p}>1$. Indeed, we have the following result.

Corollary 2. *Suppose that $g\in\mathcal{H}ol(\mathbb{D})$, $0<p<q$ and $\alpha,\beta>-1$. If $p-2<\alpha<p$ and $\frac{q(2+\alpha-p)}{p}>1$, then the following conditions are equivalent*:

(i) *I_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$.*
(ii) *M_g is a bounded operator from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$.*
(iii) $M_\infty(r,g)=\mathrm{O}((1-r)^{\frac{2+\alpha}{p}-\frac{\beta+2}{q}})$.

Corollary 2 follows immediately from Theorems 2, 3 and the following lemma.

Lemma 5. *Suppose that $g\in\mathcal{H}ol(\mathbb{D})$, $0<p<q$, $\alpha,\beta>-1$, $p-2<\alpha<p$ and $\frac{q(2+\alpha-p)}{p}>1$. If g satisfies* (26)*, then measure $\mu_{g,q,\beta}$ in $\mathbb{D}$ defined by $d\mu_{g,q,\beta}(z)=(1-|z|^2)^\beta|g'(z)|^q\,dA(z)$ is a $\frac{q}{p}(\alpha-p+2)$-Carleson measure.*

Proof. Suppose that $g \in \mathcal{H}ol(\mathbb{D})$ satisfies (26). If $\frac{2+\alpha}{p} - \frac{\beta+2}{q} > 0$, $g \equiv 0$ and the result is trivial. Hence, assume that $\frac{2+\alpha}{p} - \frac{\beta+2}{q} \leqslant 0$. Then we have that

$$M_\infty(r, g') = \mathrm{O}\big((1-r)^{\frac{2+\alpha}{p} - \frac{\beta+2}{q} - 1}\big).$$

Let $I \subset \partial\mathbb{D}$ be an interval, then

$$\mu_{g,q,\beta}\big(S(I)\big) = \int_{S(I)} \big(1-|z|^2\big)^\beta \big|g'(z)\big|^q \, dA(z) \leqslant C|I| \int_{1-\frac{|I|}{2\pi}}^1 (1-r)^{\frac{q(2+\alpha)}{p} - 2 - q} \, dr$$

$$\leqslant C|I|^{\frac{q(2+\alpha-p)}{p}}. \qquad \square$$

4. Examples

Theorem 3 gives a complete characterization of the multipliers from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ whenever $0 < p < q$, $\alpha, \beta > -1$ and $p - 2 < \alpha < p$. This characterization is very simple if either $\frac{2+\alpha}{p} > \frac{\beta+2}{q}$ or $\frac{q(2+\alpha-p)}{p} > 1$. If

$$\frac{2+\alpha}{p} \leqslant \frac{\beta+2}{q} \quad \text{and} \quad \frac{q(2+\alpha-p)}{p} \leqslant 1 \tag{28}$$

the characterization is more complicated as we cannot get rid of the condition on the measure $\mu_{g,q,\beta}$. In this section we shall be interested in this case and we shall give a number of explicit examples of functions which are multipliers from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$ for p, q, α, β satisfying (28).

Since $p < q$, we observe that (28) implies that $\alpha < p - 1$. We also note that (28) can be rewritten as

$$\frac{2+\alpha}{p} \leqslant \min\left(\frac{2+\beta}{q}, \frac{q+1}{q}\right). \tag{29}$$

We shall distinguish three cases: $q < \beta + 1$, $q > \beta + 1$ and $q = \beta + 1$.

4.1. Case A, $q < \beta + 1$

In this case (29) reduces to

$$\frac{2+\alpha}{p} \leqslant \frac{q+1}{q} < \frac{2+\beta}{q}, \tag{30}$$

or, equivalently,

$$(2+\alpha)\frac{q}{q+1} \leqslant p \quad \text{and} \quad q < \beta + 1. \tag{31}$$

If $\gamma > 0$ and $f \in \mathcal{H}ol(\mathbb{D})$, we say that f is a γ-Bloch function if

$$\|f\|_{\mathcal{B}_\gamma} \stackrel{\text{def}}{=} |f(0)| + \sup_{z\in\mathbb{D}} (1-|z|^2)^\gamma |f'(z)| < \infty.$$

The space of all γ-Bloch functions is called the γ-Bloch space and will be denoted by $\mathcal{B}_\gamma$. When $\gamma = 1$ we have the classical Bloch space $\mathcal{B}$ (see [3]).

We shall prove the following result.

Theorem 7. *Suppose that* $0 < p < q$, $\alpha, \beta > -1$, $p - 2 < \alpha$ *and* p, q, α, β *satisfy* (30). *Then*

$$\gamma < \frac{\beta+1}{q} \quad \Rightarrow \quad \mathcal{B}_\gamma \subset M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta). \tag{32}$$

Proof. Suppose that p, q, α, β are as in the statement, $\gamma < \frac{\beta+1}{q}$ and $f \in \mathcal{B}_\gamma$. Assume also, without loss of generality, that $\gamma > 1$. Let I be an interval in $\partial\mathbb{D}$. Since $\frac{q(2+\alpha-p)}{p} \leqslant 1 < \beta - q\gamma + 2$, we deduce that

$$\begin{aligned}\int_{S(I)} (1-|z|^2)^\beta |f'(z)|^q \, dA(z) &\leqslant \|f\|^q_{\mathcal{B}_\gamma} \int_{S(I)} (1-|z|^2)^{\beta-q\gamma} \, dA(z)\\ &\leqslant C\|f\|^q_{\mathcal{B}_\gamma} |I|^{\beta-q\gamma+2}\\ &\leqslant C\|f\|^q_{\mathcal{B}_\gamma} |I|^{\frac{q(2+\alpha-p)}{p}}.\end{aligned}$$

Hence the measure $\mu_{f,q,\beta}$ defined by $d\mu_{f,q,\beta}(z) = (1-|z|^2)^\beta |f'(z)|^q \, dA(z)$ is a $\frac{q(2+\alpha-p)}{p}$-Carleson measure. On the other hand, since $f \in \mathcal{B}_\gamma$,

$$M_\infty(r, f) = \mathrm{O}\left(\frac{1}{(1-r)^{\gamma-1}}\right) \quad \text{as } r \to 1^-. \tag{33}$$

Observe that the hypotheses $\gamma < \frac{\beta+1}{q}$ and $\frac{q(2+\alpha-p)}{p} \leqslant 1$ imply $\gamma - 1 \leqslant \frac{\beta+2}{q} - \frac{\alpha+2}{p}$, which, together with (33), yields

$$M_\infty(r, f) = \mathrm{O}\left(\frac{1}{(1-r)^{\frac{\beta+2}{q} - \frac{\alpha+2}{p}}}\right) \quad \text{as } r \to 1^-.$$

Using Theorem 3, we deduce that $f \in M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$. □

Theorem 7 asserts that, for the range of values p, q, α, β under consideration, the condition of being a γ-Bloch function for some $\gamma < \frac{\beta+1}{q}$ is sufficient for being a multiplier from $\mathcal{D}^p_\alpha$ to $\mathcal{D}^q_\beta$. It is natural to ask whether or not it is also necessary. Next we shall show that the answer to this question is negative in a very strong sense. Indeed, we shall construct an analytic function f which belongs to $M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$ for all those values of $p, q\ \alpha, \beta$ but not to $\bigcup_{\gamma < \frac{\beta+1}{q}} B_\gamma$.

Theorem 8. *Suppose that $\alpha, \beta > -1$ and $2 + \alpha < q + 1 < \beta + 2$. Then there exists a function $f \in \mathcal{H}ol(\mathbb{D})$ such that*

$$f \in \bigcap_{p \in [\frac{q(2+\alpha)}{q+1}, q^*)} M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta\big) \Big\backslash \bigcup_{\gamma < \frac{\beta+1}{q}} B_\gamma,$$

where $q^* \stackrel{\text{def}}{=} \min(\alpha + 2, q)$.

The following result will be used in the proof of Theorem 8. It can be proved with the arguments used in the proof of [15, Theorem 3.1], we shall omit the details.

Proposition 3. *Suppose that $0 < q < \infty$, $\beta > -1$ and $f \in \mathcal{H}ol(\mathbb{D})$, $f(z) = \sum_{k=0}^\infty a_k z^k$ $(z \in \mathbb{D})$. If*

$$\sum_{n=0}^\infty 2^{-n(\beta+1-q)} \left(\sum_{k=2^n}^{2^{n+1}-1} |a_k| \right)^q < \infty,$$

then the measure $\mu_{f,q,\beta}$ in $\mathbb{D}$ defined by $d\mu_{f,q,\beta}(z) = (1-|z|^2)^\beta |f'(z)|^q \, dA(z)$ is an 1-Carleson measure.

Proof of Theorem 8. Let q, α, and β be as in the statement. We are going to construct a function $f \in \mathcal{H}ol(\mathbb{D})$ satisfying the following conditions:

(a) The measure $\mu_{f,q,\beta}$ in $\mathbb{D}$ defined by $d\mu_{f,q,\beta}(z) = (1-|z|^2)^\beta |f'(z)|^q \, dA(z)$ is an 1-Carleson measure.
(b) $M_\infty(r, f) = \mathrm{O}((\frac{1}{1-r})^{\frac{\beta+1}{q}-1})$ as $r \to 1$.
(c) There exist $C > 0$, $s > 0$ and a sequence $\{r_n\}_{n=1}^\infty \uparrow 1$ such that

$$M_\infty(r_n, f') \geqslant C \frac{1}{(1-r_n)^{\frac{\beta+1}{q}} \log^s \frac{1}{1-r_n}}, \quad n = 1, 2, \ldots.$$

This will prove the theorem. Indeed, condition (c) gives that $f \notin \bigcup_{\gamma < \frac{\beta+1}{q}} B_\gamma$. Notice that if $\frac{q(2+\alpha)}{q+1} \leqslant p < q^* \leqslant q$ then

$$\frac{q(2+\alpha-p)}{p} \leqslant 1 \quad \text{and} \quad \frac{\beta+1}{q} - 1 = \frac{\beta+2}{q} - \frac{q+1}{q} \leqslant \frac{\beta+2}{q} - \frac{2+\alpha}{p}.$$

Thus, using Theorem 3, we see that if $f \in \mathcal{H}ol(\mathbb{D})$ satisfies (a) and (b) then

$$f \in \bigcap_{p \in [\frac{q(2+\alpha)}{q+1}, q^*)} M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta\big).$$

Hence, it remains to construct f.

Take $m \in \mathbb{N}$ and $s > 0$ such that

$$\frac{2^{m\frac{\beta+1}{q}}}{e^{2^m}} < 1, \qquad s > \frac{1}{q}. \tag{34}$$

Let

$$f(z) = \sum_{k=1}^{\infty} \frac{2^{mk(\frac{\beta+1}{q}-1)}}{k^s} z^{2^{mk}} = \sum_{j=0}^{\infty} a_j z^j, \quad z \in \mathbb{D}.$$

Since $s > \frac{1}{q}$, we have that

$$\sum_{n=0}^{\infty} 2^{-n(\beta+1-q)} \left(\sum_{j=2^n}^{2^{n+1}-1} |a_j| \right)^q = \sum_{k=1}^{\infty} 2^{-mk(\beta+1-q)} \frac{2^{mk(\beta+1-q)}}{k^{sq}} = \sum_{k=1}^{\infty} k^{-sq} < \infty,$$

so, using Proposition 3, we deduce that the measure $\mu_{f,q,\beta}$ is an 1-Carleson measure.

On the other hand,

$$\left|zf'(z)\right| \leqslant \sum_{k=1}^{\infty} \frac{2^{mk\frac{\beta+1}{q}}}{k^s} |z|^{2^{mk}} = I(z,n) + II(z,n), \tag{35}$$

where

$$I(z,n) = \sum_{k=1}^{n} \frac{2^{mk\frac{\beta+1}{q}}}{k^s} |z|^{2^{mk}}, \qquad II(z,n) = \sum_{k=n+1}^{\infty} \frac{2^{mk\frac{\beta+1}{q}}}{k^s} |z|^{2^{mk}}.$$

If $|z| = r_n \stackrel{\text{def}}{=} 1 - \frac{1}{2^{mn}}$, then

$$I(z,n) \leqslant \sum_{k=1}^{n} 2^{mk\frac{\beta+1}{q}} \leqslant C 2^{mn\frac{\beta+1}{q}} = C 2^{m\frac{\beta+1}{q}} 2^{m(n-1)\frac{\beta+1}{q}} = \widetilde{C} 2^{m(n-1)\frac{\beta+1}{q}}, \tag{36}$$

and, using the elementary inequality $(1 - \frac{1}{k})^k \leqslant e^{-1}$ $(k \geqslant 1)$ and (34), we obtain

$$\begin{aligned} II(z,n) &\leqslant n^{-s} \sum_{k=n+1}^{\infty} 2^{mk\frac{\beta+1}{q}} e^{-2^{m(k-n)}} \leqslant n^{-s} \sum_{j=1}^{\infty} 2^{m(j+n)\frac{\beta+1}{q}} e^{-2^{mj}} \\ &\leqslant n^{-s} 2^{mn\frac{\beta+1}{q}} \sum_{j=1}^{\infty} \left(\frac{2^{m\frac{\beta+1}{q}}}{e^{2^m}} \right)^j \leqslant C n^{-s} 2^{mn\frac{\beta+1}{q}} \\ &\leqslant \widetilde{C} n^{-s} 2^{m(n-1)\frac{\beta+1}{q}}. \end{aligned} \tag{37}$$

Using (35)–(37) we deduce that

$$M_\infty(r_n, f') \leqslant C 2^{m(n-1)\frac{\beta+1}{q}} = C\left(\left(\frac{1}{1-r_{n-1}}\right)^{\frac{\beta+1}{q}}\right).$$

Bearing in mind that the functions

$$r \mapsto M_\infty(r, f') \quad \text{and} \quad r \mapsto \left(\frac{1}{1-r}\right)^{\frac{\beta+1}{q}}$$

are increasing in $(0, 1)$, it follows easily that

$$M_\infty(r, f') = \mathrm{O}\left(\left(\frac{1}{1-r}\right)^{\frac{\beta+1}{q}}\right) \quad \text{as } r \to 1.$$

Since $\frac{\beta+1}{q} > 1$, this implies that

$$M_\infty(r, f) = \mathrm{O}\left(\left(\frac{1}{1-r}\right)^{\frac{\beta+1}{q}-1}\right) \quad \text{as } r \to 1,$$

that is, f satisfies (b).

It remains to prove that f satisfies (c). Setting $r_n = 1 - \frac{1}{2^{mn}}$ $(n \geqslant 1)$ as above, we see that

$$r_n M_\infty(r_n, f') \geqslant r_n f'(r_n) \geqslant \sum_{k=1}^{n} \frac{2^{mk\frac{\beta+1}{q}}}{k^s} r_n^{2^{mk}} \geqslant \sum_{k=1}^{n} \frac{2^{mk\frac{\beta+1}{q}}}{k^s} r_n^{2^{mn}} \geqslant 4^{-1} n^{-s} \sum_{k=1}^{n} 2^{mk\frac{\beta+1}{q}}$$
$$\geqslant C n^{-s} 2^{mn\frac{\beta+1}{q}} \geqslant C \frac{1}{(1-r_n)^{\frac{\beta+1}{q}} \log^s \frac{1}{1-r_n}}.$$

This finishes the proof. □

4.2. Case B, $q > \beta + 1$

In this case (29) reduces to

$$\frac{2+\alpha}{p} \leqslant \frac{2+\beta}{q} < \frac{q+1}{q}, \tag{38}$$

or, equivalently,

$$\frac{q(2+\alpha)}{2+\beta} \leqslant p \quad \text{and} \quad q > \beta + 1. \tag{39}$$

We note also that the condition $p < q$ implies that $\alpha < \beta$.

Since $H^\infty \subset \mathcal{B}$, Theorem 7 implies that, in Case A, $H^\infty \subset M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$. This does not remain true in Case B. Our work in this case will be mainly devoted to determining whether or not the atomic singular inner function

$$S_{\gamma,\eta}(z) = \exp\left(\gamma \frac{z+\eta}{z-\eta}\right), \quad z \in \mathbb{D},$$

where $\gamma > 0$ and $\eta \in \partial\mathbb{D}$, belongs to $M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$. The following result was proved by Jevtić in [18] and it can also be deduced from the estimates on the integral means $I_q(r, S^{(n)}_{\gamma,\eta})$ obtained in [22].

Proposition B. *Let β and q be such that $-1 < \beta < q - 1$. Then $S_{\gamma,\eta} \in \mathcal{D}^q_\beta$ if and only if*

$$2(\beta - q) + 3 > 0.$$

Since $\mathcal{D}^p_\alpha$ contains the constant functions, Proposition B implies the following:

$$\textit{If} \quad -1 < \beta < q - 1 \quad \textit{and} \quad 2(\beta - q) + 3 \leqslant 0 \quad \textit{then} \quad S_{\gamma,\eta} \notin M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta\big).$$

Let us consider now the case $-1 < \beta < q - 1$ and $2(\beta - q) + 3 > 0$. We shall prove the following result.

Theorem 9. *Suppose that $-1 < \alpha < \beta < q - 1$, $2(\beta - q) + 3 > 0$ and $p - 2 < \alpha$. Then*:

(a) *if $2\beta - q + 3 \leqslant \alpha + 2$ and $\frac{q(\alpha+2)}{\beta+2} \leqslant p < q$, then $S_{\gamma,\eta} \notin M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$;*
(b) *if $2\beta - q + 3 > \alpha + 2$ and $\frac{q(\alpha+2)}{\beta+2} \leqslant p < \frac{q(\alpha+2)}{2\beta-q+3}$, then $S_{\gamma,\eta} \notin M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$;*
(c) *if $2\beta - q + 3 > \alpha + 2$ and $\frac{q(\alpha+2)}{2\beta-q+3} \leqslant p < q$, then $S_{\gamma,\eta} \in M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$.*

Proof. Let p, q, α, β be as in the statement. Let J be an interval in $\partial\mathbb{D}$ centered at η with $|J| \leqslant \frac{1}{2}$.

For $n = 1, 2, \ldots$, set

$$J_n = \big\{e^{i\theta}\colon \arg\eta - |J|2^{-n} < \theta < \arg\eta + |J|2^{-n}\big\}, \qquad E_n = S(J_n).$$

We have

$$\int_{S(J)} \big(1 - |z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q \, dA(z) = \sum_{n=1}^{\infty} \int_{E_n \setminus E_{n+1}} \big(1 - |z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q \, dA(z). \tag{40}$$

Now we observe that

$$E_n \setminus E_{n+1} = A_n \,\dot\cup\, B_n, \tag{41}$$

where $\dot\cup$ denotes the disjoint union, and

$$A_n = \left\{ z = |z| e^{i\theta} \colon \frac{|J|}{2^n (2\pi)} < 1 - |z| \leqslant \frac{|J|}{2^{n-1}(2\pi)},\ |\theta - \arg \eta| < \frac{|J|}{2^n} \right\}, \tag{42}$$

$$B_n = \left\{ z = |z| e^{i\theta} \colon 1 - |z| \leqslant \frac{|J|}{2^n (2\pi)},\ \frac{|J|}{2^{n+1}} \leqslant |\theta - \arg \eta| < \frac{|J|}{2^n} \right\}. \tag{43}$$

Let us recall that

$$\begin{aligned} &\big(1 - |z|\big)^2 + \frac{2}{\pi^2} (\theta - \arg \eta)^2 \leqslant |z - \eta|^2 \leqslant \big(1 - |z|\big)^2 + (\theta - \arg \eta)^2, \\ &\frac{1}{2} \leqslant |z| < 1. \end{aligned} \tag{44}$$

Using (44) and the definitions of B_n and $S_{\gamma,\eta}$, we deduce that

$$\begin{aligned} &\int_{B_n} \big(1 - |z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q \, dA(z) \\ &\quad \geqslant C \int_{1 - \frac{|J|}{2^n (2\pi)}}^{1} \int_{\frac{|J|}{2^{n+1}} \leqslant |\theta - \arg \eta| < \frac{|J|}{2^n}} (1 - r)^\beta \exp\left(\frac{-q\gamma (1 - r^2)}{|re^{i\theta} - \eta|^2} \right) \frac{(2\gamma)^q}{|re^{i\theta} - \eta|^{2q}} \, d\theta \, dr \\ &\quad \geqslant C \int_{1 - \frac{|J|}{2^n (2\pi)}}^{1} \int_{\frac{|J|}{2^{n+1}} \leqslant |\theta - \arg \eta| < \frac{|J|}{2^n}} (1 - r)^\beta \exp\left(\frac{-q\gamma (1 - r^2)}{\frac{2}{\pi^2} \left(\frac{|J|}{2^{n+1}}\right)^2} \right) \frac{(2\gamma)^q}{\left(2\frac{|J|}{2^n}\right)^{2q}} \, d\theta \, dr \\ &\quad \geqslant C \left(\frac{|J|}{2^n} \right)^{1-2q} \int_{1 - \frac{|J|}{2^n (2\pi)}}^{1} (1 - r)^\beta \exp\left(\frac{-4q\pi^2 \gamma (1 - r)}{\left(\frac{|J|}{2^n}\right)^2} \right) dr \\ &\quad \geqslant C \left(\frac{|J|}{2^n} \right)^{3+2(\beta-q)} \int_0^{\frac{2^{n+1} q\gamma\pi}{|J|}} s^\beta \exp(-s) \, ds \\ &\quad \geqslant C \left(\int_0^{\frac{4q\gamma\pi}{|J|}} s^\beta \exp(-s) \, ds \right) \left(\frac{|J|}{2^n} \right)^{3+2(\beta-q)}. \end{aligned} \tag{45}$$

Using (40)–(43) and (45), we deduce that

$$\begin{aligned} \int_{S(J)} \big(1 - |z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q \, dA(z) &\geqslant C \int_0^{\frac{4q\gamma\pi}{|J|}} s^\beta \exp(-s) \, ds \sum_{n=1}^{\infty} \left(\frac{|J|}{2^n} \right)^{3+2(\beta-q)} \\ &\geqslant C |J|^{3+2(\beta-q)} \int_0^{\frac{4q\gamma\pi}{|J|}} s^\beta \exp(-s) \, ds. \end{aligned}$$

Hence,

$$\int_0^{\frac{4q\gamma\pi}{|J|}} s^\beta \exp(-s)\,ds \leqslant C|J|^{-(3+2(\beta-q))} \int_{S(J)} \big(1-|z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q\,dA(z). \tag{46}$$

Also, using (44) and the definitions of A_n and $S_{\gamma,\eta}$, we obtain

$$\begin{aligned}
&\int_{A_n} \big(1-|z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q\,dA(z)\\
&\leqslant \int_{1-\frac{|J|}{2^{n-1}(2\pi)}}^{1-\frac{|J|}{2^n(2\pi)}} \int_{|\theta-\arg\eta|<\frac{|J|}{2^n}} (1-r)^\beta \exp\left(\frac{-q\gamma(1-r^2)}{|re^{i\theta}-\eta|^2}\right)\frac{(2\gamma)^q}{|re^{i\theta}-\eta|^{2q}}\,d\theta\,dr\\
&\leqslant C\int_{1-\frac{|J|}{2^{n-1}(2\pi)}}^{1-\frac{|J|}{2^n(2\pi)}} \int_{|\theta-\arg\eta|<\frac{|J|}{2^n}} (1-r)^\beta \exp\left(\frac{-q\gamma(1-r^2)}{2(\frac{|J|}{2^{n-1}})^2}\right)\frac{(2\gamma)^q}{\big(\frac{|J|}{2^n}\big)^{2q}}\,d\theta\,dr\\
&\leqslant C\left(\frac{|J|}{2^n}\right)^{1-2q}\int_{1-\frac{|J|}{2^{n-1}(2\pi)}}^{1-\frac{|J|}{2^n(2\pi)}} (1-r)^\beta \exp\left(\frac{-q\gamma(1-r)}{8\big(\frac{|J|}{2^n}\big)^2}\right)dr\\
&\leqslant C\left(\frac{|J|}{2^n}\right)^{3+2(\beta-q)}\int_{\frac{2^{n-1}q\gamma}{8\pi|J|}}^{\frac{2^n q\gamma}{8\pi|J|}} s^\beta\exp(-s)\,ds\\
&\leqslant C\left(\frac{|J|}{2^n}\right)^{3+2(\beta-q)}\int_0^\infty s^\beta\exp(-s)\,ds\\
&\leqslant C\left(\frac{|J|}{2^n}\right)^{3+2(\beta-q)}.
\end{aligned}\tag{47}$$

In a similar way we can prove that

$$\int_{B_n} \big(1-|z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q\,dA(z) \leqslant C\left(\frac{|J|}{2^n}\right)^{3+2(\beta-q)}. \tag{48}$$

Suppose that α, β, p, q are in the conditions of either (a) or (b) then $\frac{q}{p}(\alpha+2-p) - (3+2(\beta-q)) > 0$ which, using (46) with $|J|$ arbitrarily small, implies that the measure

$(1-|z|)^\beta |S'_{\gamma,\eta}(z)|^q\, dA(z)$ is not a $\frac{q}{p}(\alpha+2-p)$-Carleson measure. Then Theorem 3 implies that $S_{\gamma,\eta} \notin M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$.

Suppose now that α, β, p, q are in the conditions of (c). Then $0 < \frac{q}{p}(\alpha+2-p) \leqslant 3+2(\beta-q)$ and, using (40)–(43), (47) and (48), we deduce that

$$\int_{S(J)} \big(1-|z|\big)^\beta \big|S'_{\gamma,\eta}(z)\big|^q\, dA(z) \leqslant C \sum_{n=1}^{\infty} \left(\frac{|J|}{2^n}\right)^{3+2(\beta-q)} \leqslant C|J|^{3+2(\beta-q)} \leqslant C|J|^{\frac{q(\alpha+2-p)}{p}}.$$

Hence, the measure $(1-|z|)^\beta |S'_{\gamma,\eta}(z)|^q\, dA(z)$ is a $\frac{q}{p}(\alpha+2-p)$-Carleson measure, which using Theorem 3 and the fact that $S_{\gamma,\eta} \in H^\infty$ shows that $S_{\gamma,\eta} \in M(\mathcal{D}^p_\alpha, \mathcal{D}^q_\beta)$. □

4.3. Case C, $q = \beta + 1$

In this case (29) reduces to

$$\frac{2+\alpha}{p} \leqslant \frac{q+1}{q} = \frac{\beta+2}{q}.$$

Hence, we are dealing with the multipliers $g \in M(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1})$ with

$$p, q > 0, \qquad \alpha > -1 \quad \text{and} \quad \frac{q(2+\alpha)}{q+1} \leqslant p < q^* \stackrel{\text{def}}{=} \min(q, 2+\alpha). \tag{49}$$

We have the following result for $q \geqslant 2$.

Theorem 10. *Suppose that p, q, α satisfy* (49) *and $2 \leqslant q < \infty$. Then*

$$\bigcap_{p \in [\frac{q(2+\alpha)}{q+1}, q^*)} M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1}\big) = H^\infty.$$

Proof. Lemma 1 shows that

$$\bigcap_{p \in [\frac{q(2+\alpha)}{q+1}, q^*)} M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1}\big) = M\big(\mathcal{D}^{\frac{q(2+\alpha)}{q+1}}_\alpha, \mathcal{D}^q_{q-1}\big). \tag{50}$$

Hence, we have to prove that $M(\mathcal{D}^{\frac{q(2+\alpha)}{q+1}}_\alpha, \mathcal{D}^q_{q-1}) = H^\infty$. Now, Theorem 3 gives that the following conditions are equivalent:

(i) $g \in M(\mathcal{D}^{\frac{q(2+\alpha)}{q+1}}_\alpha, \mathcal{D}^q_{q-1})$;
(ii) $g \in H^\infty$ and the measure $\mu_{g,q,q-1}$ in $\mathbb{D}$ defined by

$$d\mu_{g,q,q-1}(z) = \big(1-|z|^2\big)^{q-1} \big|g'(z)\big|^q$$

is an 1-Carleson measure.

Now, recalling that $H^\infty \subset \mathcal{BMOA} \subset \mathcal{B}$, using the characterization of $\mathcal{BMOA}$ in terms of Carleson measures (see, e.g., [6,12,14]) and bearing in mind that $q \geqslant 2$, we easily see that the condition $g \in H^\infty$ implies that $\mu_{g,q,q-1}$ is an 1-Carleson measure. Thus (ii) is equivalent to saying that $g \in H^\infty$. □

It is natural to ask whether or not Theorem 10 remains true for $0 < q < 2$. The answer to this question is negative. Indeed, Vinogradov proved in [25, Theorem 3.11] that there exists a Blaschke product B such that $B \notin \bigcup_{0<q<2} \mathcal{D}^q_{q-1}$. Since the constants functions belong to all the spaces $\mathcal{D}^p_\alpha$, it follows that

$$H^\infty \not\subset M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1}\big), \quad p > 0,\ \alpha > -1,\ 0 < q < 2.$$

Using the main result in [13] we can also say that

$$\mathcal{A} \not\subset M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1}\big), \quad p > 0,\ \alpha > -1,\ 0 < q < 2,$$

where $\mathcal{A}$ denotes the disc algebra, that is the space of all $f \in \mathcal{H}ol(\mathbb{D})$ that have a continuous extension to the closed unit disc $\overline{\mathbb{D}}$.

On the other hand, noticing that (50) and the equivalence of the conditions (i) and (ii) stated in the proof of Theorem 10 remain true for $0 < q < 2$, and using [25, Theorems 2.7 and 2.2] we deduce that the atomic function $S_{\gamma,\eta}$ ($\gamma > 0, \eta \in \partial\mathbb{D}$,) satisfies that

$$S_{\gamma,\eta} \in \bigcap_{p \in [\frac{q(2+\alpha)}{q+1}, q^*)} M\big(\mathcal{D}^p_\alpha, \mathcal{D}^q_{q-1}\big).$$

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